AI-Powered Criminal Network Analysi

An end-to-end pipeline that turns fra
records, financial transactions, and so
identifies key influencers, flags suspici
investigator dashboard.

**Built as a working proof of concept
not just written.**

The Core Result

Using generated synthetic data, one p

written police report — but the syste

by fusing pho

data source alone would have found

person in the network

fusion finds hidden connections tha

Quick Start

```
# 1. Install dependencies
pip install spacy rapidfuzz
python -m spacy download en_

# 2. Run the full pipeline
python3 step1_generate_data.
python3 step1b_generate_soc
python3 step2_extract_entit
```

```
python3 step3_entity_resolu
python3 step4_extract_relat
python3 step5_build_graph.py
python3 step6_graph_analyti
python3 step7_anomaly_detec
python3 step8_export_dashbo
python3 step9_build_dashboa

# 3. View the results
open dashboard/index.html

# 4. (Optional) Run the live
python -m uvicorn step10_api
```

Folder Structure

```
crime_network/
├── step1_generate_data.py
├── step1b_generate_social.py
├── step2_extract_entities.py
├── step3_entity_resolution.py
├── step4_extract_relations.py
├── step5_build_graph.py
├── step6_graph_analytics.py
├── step7_anomaly_detection.py
├── step8_export_dashboard.py
├── step9_build_dashboard.py
├── step10_api.py
├── data/
│   ├── firs.txt, cdr.csv, f
│   ├── social_posts.csv, so
│   ├── extracted_entities.j
```

```
%   %%% graph_full.pkl, grap
%   %%% centrality_scores.js
%%% dashboard/
    %%% dashboard_data.json
    %%% index.html
```

Rule of thumb: . files live in the dashboard/ .

Pipeline Stages Explained

#	Stage	What It D
1	Data Generation	Creates re records, fi

1b	Generation	Creates fa friend/foll
2	Entity Extraction	Pulls nam of police r
3	Entity Resolution	Matches n one canon
4	Relationship Extraction	Turns co- transactio
5	Graph Construction	Builds the
6	Graph Analytics	Scoreswh sub-group

7	Anomaly Detection	Flags circu frequency
8	Data Export	Consolida computes
9	Dashboard	Interactive UI
10	Backend API	Live query

Known Limitations (Be Upfront About

Synthetic data only. All names, recor individuals, cases, or data are involved **NER isn't perfect.** Off-the-shelf spaC testing) — a real system would need

extractions.

Anomaly detection needed tuning. false positives on a 5-person graph be a genuine, documented lesson: naive

Centrality results are illustrative, no the "hidden" person until a third data no single metric or data source tells th

Scores are signals for human review guilty of anything. High centrality or a not "this person did something wrong

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Tech Stack

NLP: spaCy (`en_core_web_sm`)

Entity resolution: RapidFuzz

Graph & analytics: NetworkX (betwe

Backend: FastAPI

Frontend: Vanilla HTML/CSS/JS (SVG

Visualization (dev-side): Matplotlib

Extending This Project

Add more data sources (surveillance r
pattern as Step 1b.

Swap NetworkX for Neo4j if you need

Fine-tune a custom NER model on rea
general-purpose spaCy model.

Add authentication/RBAC to `step10`